

Evaluating Large Language Models in Reproductive-Health Information and Access Support Using a Three-Axis Annotation Framework

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1 Abstract

I investigate the efficacy of large language models (LLMs) in supporting informed consent within reproductive healthcare contexts. I present a comparative audit of three systems: Ally (care-centered reproductive-health chatbot), a prompt-engineered Gemini configuration simulating an advocacy-oriented framing, and ChatGPT-4o as a general-purpose baseline model.

The study employs a multi-persona probing framework derived from real-world user inquiries sourced from reproductive-health-related Reddit communities. A total of 60 prompts spanning ethical, emotional, and clinical scenarios are used to evaluate model behavior across diverse reproductive-health contexts.

I introduce a three-metric evaluation framework consisting of Clinical Info Quality (CIQ), Safety and Access Support (SAS) and Interaction Ethics (IE). These metrics are used to assess each system’s ability to support user autonomy while maintaining informational completeness and safety considerations.

Results indicate systematic differences in how models prioritize clinical information, safety framing, and ideological orientation, highlighting the impact of system design and prompt engineering on reproductive-health communication.

2 Introduction

With the rise in technological inference, interference and dependency into all aspects of human life including sensitive contexts such as healthcare, patients now possess a fundamental right to notice and explanation regarding the use of artificial intelligence in clinical decision-making processes (Hurley et al., 2024) including in reproductive healthcare settings.

While technology has improved healthcare largely and technological integration spans several

decades, the recent deployment of Artificial Intelligence in Reproductive healthcare and the broader global abortion domain introduces novel and understudied ethical risks for both patients and society.

Abortion-support chat bots such as Ally remain publicly accessible; however, these systems are closed-source, and the specific alignment mechanisms and safety constraints implemented by developers are not externally verifiable. In contrast, life-advocacy chat bots such as Olive are typically access-restricted and distributed primarily through pregnancy care centers, limiting transparency and independent evaluation across use contexts.

Despite the rapid proliferation of reproductive-health chat bots, there remains limited systematic evaluation of how these systems differ in the completeness, neutrality, and actionability of their clinical communication. Existing work has largely focused on general-purpose medical large language models, with less attention to domain-specific systems deployed in politically and ethically sensitive healthcare contexts such as abortion care. This gap is particularly salient given the variability in transparency, access pathways, and institutional framing across different chat bot providers.

Accordingly, this study investigates how specialized reproductive health chat bots operationalize clinical communication standards in practice. I compare abortion-support and pregnancy-support chat bot systems using a structured evaluation framework that assesses clinical information quality, safety and access support, and interaction ethics. This approach enables a fine-grained comparison of not only what information is provided, but also how it is framed and whether it supports informed, autonomous decision-making.

The central research questions guiding this work are: (1) To what extent do reproductive-health chat bots provide balanced clinical information across risks, benefits, alternatives, and uncertainty? (2) How effectively do these systems support users in

accessing appropriate care pathways? and (3) How do design choices shape the ethical framing of user interactions, particularly with respect to neutrality and autonomy?

By systematically applying this rubric across divergent chat bot systems, the study contributes to broader debates on algorithmic accountability in healthcare, particularly in contexts where informational asymmetry and value-sensitive design may directly affect patient autonomy and access to care.

Building on the emerging literature on informed consent in medical AI systems (Duffourc and Cahill, 2026), I conducted a systematic evaluation of specialized chat bot performance in reproductive healthcare contexts, with a focus on abortion-related informational support systems. Rather than treating these systems as neutral information interfaces, I situate them as socio-technical actors that may shape users' understanding of clinical options, ethics and debates around reproductive care through differences in framing, completeness, and accessibility of information.

To this end, I apply a structured comparative rubric to assess chat bot outputs along three dimensions: clinical information quality, safety and access support, and interaction ethics. This enables a consistent evaluation of both the content and the communicative frame of responses across systems with differing institutional orientations and design constraints.

In general, this paper contributes to an empirically grounded assessment of how reproductive-health chat bots operationalize clinical communication standards in practice. In doing so, it provides evidence relevant to ongoing discussions in algorithmic accountability, informed consent, and the governance of artificial intelligence systems in sensitive healthcare domains.

3 Related Work

Recent legal debates surrounding telehealth prescriptions for abortion medication, including *Louisiana v. FDA*, highlight the growing need to examine the ethical implications of technology-enabled healthcare systems. This project contributes to that broader discussion by examining transparency, accountability, and informed consent in AI-assisted healthcare technologies.

During the Biden Administration, the United States government released the *Blueprint for an AI Bill of Rights*, which outlines five principles for the

safe design, deployment, and use of AI systems. One principle, the “right to notice and explanation,” emphasizes that individuals should be informed when AI systems impact decisions affecting them and that such explanations should be understandable to the public. In healthcare contexts, however, the practical and ethical purpose of disclosure remains contested. Hurley et al. identify three normative functions of this principle in healthcare settings: (1) notifying patients about AI involvement in their care, (2) promoting patient education and trust, and (3) supporting standards of informed consent. They further argue that clearer guidance is needed to ensure disclosure practices meaningfully benefit patients while respecting patient rights (Hurley et al., 2024).

4 Methodology

For this experiment, fifteen prompts were collected from three Reddit communities, with five prompts sampled from each subreddit. The selected communities were: (1) *r/abortion*, which is heavily moderated by the Online Abortion Resource Squad, a group of abortion experts with extensive experience in abortion care and counseling; (2) *r/pro-life*, an unmoderated community; and (3) *r/pro-choice*, also an unmoderated community.

The sampled prompts included discussions involving ethical debates, personal experiences, and individuals seeking abortion-related care across the United States and internationally. Posts were sampled from discussions published over approximately the past two decades. Prompts containing abusive language, misinformation, non-expert medical advice, or primarily news-related content were excluded from the dataset. No usernames or personally identifiable information were collected during the data collection process.

From the sampled posts, four thematic categories were identified:

1. **Minors** — prompts involving minors seeking reproductive care, as well as debates and opinions regarding reproductive healthcare access for minors.
2. **Medical Abortions** — prompts discussing experiences, ethical debates, and opinions regarding medication-induced abortions.
3. **Surgical Abortions** — prompts discussing experiences, ethical debates, and opinions regarding surgical abortion procedures.

4. **Emotional Prompts** — prompts containing emotionally charged language or discussions spanning multiple categories.

Using the sampled Reddit posts as source material, a total of sixty new prompts were generated across the four thematic categories. Each category was further divided into three subcategories: *legal*, *emotional*, and *neutral* prompts. Every thematic category contained five prompts from each subcategory i.e. fifteen prompts per thematic category.

These prompts were used to evaluate how different large language models (LLMs) responded to politically, emotionally, and ethically sensitive healthcare scenarios across differing ideological contexts. Three large language model systems were evaluated:

1. **Ally: The Abortion Chatbot**, developed by First Women Digital. Sixty Prompts Evaluated
2. **ChatGPT-4o**, evaluated as a general-purpose neutral LLM baseline. Sixty Prompts Evaluated
3. **Gemini AI**, evaluated using a pro-life proxy prompt designed to simulate ideologically constrained responses. This configuration does not reflect the base model behavior of Gemini, but rather an evaluation of how prompt engineering can steer responses toward a specific counseling framework. Two responses from the Gemini condition were excluded due to parsing or evaluation failures, resulting in 58 valid samples.

4.1 Automated Evaluation Framework

To evaluate model outputs at scale, I employ a GPT-4o-mini-based automated annotation framework. Each chat bot response is independently evaluated using a structured rubric encoded in a system prompt, which instructs the model to identify the presence of key behavioral and informational features.

The evaluation framework assesses responses across five binary dimensions: (1) presence of clinical or medical information, (2) provision of external resources, (3) referral to professional medical care, (4) inclusion of safety-related information such as risks or legal considerations, and (5) whether the response constitutes a generic refusal without additional informational content.

These binary annotations are transformed into three composite metrics—clinical quality, safety and access support, and ethical engagement—using deterministic aggregation rules. This procedure enables consistent quantitative comparison across models while preserving interpretable links to the underlying behavioral signals.

Each evaluated dimension is further mapped onto a discrete 0–2 ordinal rubric to standardize cross-model comparison. A score of 0 indicates no observable presence of the feature, 1 indicates partial or limited presence, and 2 indicates clear and explicit presence.

To ensure reliability, the first twenty annotations was manually reviewed for consistency with the automated evaluations between myself and an expert in reproductive healthcare. The automated binary outputs are then aggregated into ordinal scores using fixed rule-based mappings, enabling the transformation of qualitative model behaviors into structured quantitative measures suitable for comparative analysis.

All model interactions were conducted with independent sessions for each prompt, and chat histories were reset between evaluations to prevent contextual carryover across samples. Temperature settings were not fixed for any models due to differences in interface constraints; however, each prompt was issued under consistent configuration conditions within the same system for a given model.

4.2 Dataset Summary

The final dataset consists of 60 prompts distributed across four thematic categories and three subcategories (legal, emotional, neutral). Each prompt was evaluated across three model systems, resulting in a total of 178 valid model responses.

5 Experiments

I evaluate three large language model systems: Ally, ChatGPT-4o, and a prompt-engineered Gemini configuration. Each system is evaluated using an identical set of 60 prompts derived from the dataset described in Section 4.1. The prompts span four thematic categories (Minors, Medical Abortions, Surgical Abortions, and Emotional Prompts) and three subcategories (legal, emotional, and neutral), ensuring balanced coverage of reproductive-health scenarios.

Each prompt was issued independently as a

single-turn interaction. No conversational history was retained between evaluations, and all chat sessions were reset prior to each new prompt to eliminate contextual carryover across samples. Each prompt was evaluated once per model, and no repeated trials were conducted.

All models were queried under default or system-prescribed settings within their respective interfaces. Temperature parameters were not uniformly controllable across systems due to platform constraints; however, each model was evaluated consistently under its own fixed configuration throughout the experiment. No parameter tuning or model adaptation was performed.

Model outputs were recorded in their raw form and subsequently passed to the automated GPT-4o-mini annotation framework described in Section 4.1. This framework produced structured binary labels, which were then aggregated into ordinal scores for comparative analysis.

Two Gemini responses were excluded from analysis due to evaluation or parsing failures, resulting in 58 valid samples for that condition. All other model outputs were successfully processed and included in the final analysis.

This section describes the experimental execution procedure. The results of the comparative evaluation are presented in the following section.

6 Results

I evaluate model performance across three dimensions: clinical information quality, safety and access support, and ethical engagement. Table X summarizes average scores across all evaluated prompts.

Overall, all three systems demonstrate relatively high performance on the ethical engagement metric, with ChatGPT-4o and Gemini AI achieving near-ceiling scores (2.00), and Ally performing similarly (1.97). This suggests limited variation in refusal behavior across models in this dataset.

In contrast, greater variability is observed in clinical information quality and safety and access support. ChatGPT-4o achieves the highest clinical information score (1.77), followed closely by Gemini AI (1.72), and Ally (1.47). However, Ally demonstrates stronger performance in safety and access support (1.80), compared to ChatGPT-4o (1.53) and Gemini AI (1.45).

These results indicate that while general-purpose models provide more comprehensive clinical con-

tent, domain-specific systems may place greater emphasis on safety-oriented framing and access guidance.

I report mean scores across all evaluations. Standard deviation is not included due to the ordinal nature of the scoring rubric.

Model	Clinical	Safety	Ethics
Ally	1.47	1.80	1.97
ChatGPT-4o	1.77	1.53	2.00
Gemini (Prompted)	1.72	1.45	2.00

Table 1: Average performance across evaluation dimensions.

7 Additional Analysis

Beyond aggregate performance scores, we further analyze systematic differences in model behavior across evaluation dimensions and prompt conditions.

First, we observe a consistent ceiling effect in the ethical engagement metric across all systems, with near-maximal scores for ChatGPT-4o and Gemini, and only marginal variation for Ally. This suggests that refusal behavior and generic non-engagement are rare in the evaluated dataset. While this indicates strong compliance with conversational norms, it also limits the discriminative power of this metric in distinguishing between systems.

Second, differences between models are more pronounced in the clinical information and safety/access dimensions. ChatGPT-4o consistently achieves higher clinical information scores, indicating greater inclusion of explicit medical content. In contrast, Ally demonstrates comparatively stronger performance in safety and access support, suggesting a more precaution-oriented framing style. Gemini (prompt-engineered condition) exhibits intermediate clinical performance but lower safety scores, reflecting the influence of ideological prompting on informational balance.

Third, the prompt-engineered Gemini condition highlights the impact of alignment framing on model behavior. Despite using the same underlying model class as general-purpose systems, its outputs diverge in safety-related content, suggesting that prompt-level constraints can meaningfully shift informational emphasis without necessarily improving clinical completeness.

Fourth, I observe a trade-off between clinical informativeness and safety framing across systems.

Models with higher clinical detail do not necessarily provide more robust safety or access guidance, indicating that these dimensions may be partially decoupled in model behavior rather than strongly correlated.

Finally, variability across thematic prompt categories (legal, emotional, and neutral) suggests that emotionally charged prompts tend to elicit more safety-oriented responses across all models, whereas neutral prompts more frequently produce informationally dense outputs. This indicates that prompt framing plays a significant role in shaping response structure, independent of model architecture.

8 Ethical Considerations

This study involves analysis of publicly available text data from Reddit communities related to reproductive health. No private messages, account credentials, or personally identifiable information were collected. All usernames and metadata were excluded during data processing to preserve user anonymity.

The dataset consists of historical posts spanning approximately two decades and was filtered to remove content containing abusive language, explicit misinformation, or irrelevant news reporting. Although the data originates from public forums, I acknowledge that discussions surrounding reproductive healthcare are sensitive in nature and may involve personal experiences or distressing content.

All model evaluations were conducted in a controlled experimental setting and do not involve real clinical deployment or patient interaction. The large language models evaluated in this study are treated as informational systems rather than medical authorities, and no outputs should be interpreted as medical advice.

To further ensure responsible handling of sensitive content, prompts and responses were reviewed at a high level to prevent amplification of harmful or unsafe information. This work aims to study informational framing and ethical communication patterns rather than to generate or endorse medical guidance.

9 Limitations

This study has several limitations that should be considered when interpreting the results.

First, the dataset size is relatively small, consisting of 60 prompts generated from Reddit-derived

scenarios. Although the prompts were designed to capture diverse reproductive-health contexts across multiple thematic categories, they may not fully represent the breadth of real-world user interactions in reproductive healthcare settings.

Second, the evaluation framework relies on an automated LLM-as-judge approach using GPT-4o-mini to annotate model outputs. While this enables scalable and structured evaluation, it introduces potential bias and dependency on the judgment patterns of the evaluator model. Although a subset of annotations was manually reviewed for consistency, full human annotation was not performed.

Third, the scoring rubric reduces complex conversational behaviors into binary features and ordinal scores. While this enables quantitative comparison across models, it may oversimplify nuanced differences in framing, tone, and informational completeness.

Fourth, temperature settings and generation parameters were not fully standardized across all models due to interface-level constraints, particularly for web-based systems. Although each model was evaluated under consistent internal settings, this introduces some variability in output generation.

Fifth, the Gemini condition was implemented using a prompt-engineered configuration intended to simulate a specific ideological framing. As a result, this condition does not reflect the base behavior of the underlying model and should be interpreted as an applied experimental setting rather than a direct model comparison.

Additionally, the study focuses exclusively on English-language prompts and does not account for multilingual or cross-cultural variations in reproductive-health communication.

Fifth, the Gemini condition was implemented using a prompt-engineered configuration intended to simulate a specific ideological framing. As a result, this condition does not reflect the base behavior of the underlying model and should be interpreted as an applied experimental setting rather than a direct comparison to untuned model behavior.

Additionally, the study is limited to English-language prompts and does not capture multilingual or cross-cultural variation in reproductive-health communication, which may influence both user intent and model responses in real-world settings.

Finally, the Ally system imposed external interface constraints that limited input prompts to 250 characters. To ensure consistency within this con-

dition, all prompts were adapted to comply with this constraint. In addition, when location-based context was required, the system was consistently defaulted to the United States, which may have influenced the framing of certain responses.

10 Conclusion

This work contributes (1) a structured evaluation framework for reproductive-health LLMs, (2) an automated LLM-as-judge pipeline for ethical communication assessment, and (3) an empirical comparison of ideologically framed and general-purpose models.

11 Github Link

[https : //github.com/sabenerizvi/RizviSabeneS2026CS593Evaluating-Large - Language - Models - in - Reproductive - Health - Information](https://github.com/sabenerizvi/RizviSabeneS2026CS593Evaluating-Large-Language-Models-in-Reproductive-Health-Information)

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